

Chapter 8 — Evaluation and Management of Haematuria

- Haematuria has been recognised as a sign of illness since antiquity and is now one of the commonest indications for urological evaluation.
- Up to 65% of patients with haematuria have a urological or renal condition, many needing treatment — hence the differential diagnosis, principles of evaluation and management strategies must be known.
- Presentation spans asymptomatic microhaematuria to refractory haemorrhagic cystitis, with implications for health-system resources and difficult clinical decisions.
- Knowing the risk factors for genitourinary malignancy in microscopic haematuria allows timely, appropriate evaluation; severe upper and lower tract bleeding is often life-threatening in elderly, comorbid patients.

CLASSIFICATION AND TIMING OF HAEMATURIA

- Classified by visibility, timing within the urinary stream, and associated symptoms.
- Gross haematuria (GH) — also frank, macroscopic or visible haematuria — is visible to the naked eye. Its timing within the stream suggests the source:

Timing of GH	Usual source
Initial	Urethra
Terminal	Bladder trigone, bladder neck or prostate
Total	Bladder or upper urinary tract
Cyclic (female)	Gynaecological cause (e.g. endometriosis)

- **Pigmenturia must be excluded:** endogenous pigments (bilirubin, myoglobin, porphyrins), foods (beetroot, rhubarb), drugs (phenazopyridine) or dehydration — distinguished by urinalysis (UA) with microscopy.
- **False-positive dipstick for haemoglobin:** myoglobinuria, semen and bacterial peroxidases → microscopy is required to confirm haematuria.
- **Exclude vaginal bleeding:** careful menstrual history, specimen collected outside menstrual or gynaecological bleeding, or, if necessary, a catheterised specimen. In incontinent patients GH may present as blood spotting on undergarments.
- Once vaginal bleeding and mimics are excluded, a urological source must be suspected.
- **Microhaematuria (MH) is a sign, not a symptom:** a laboratory diagnosis of red blood cells (RBCs) on urine microscopy, not visible to the eye.
- **Prevalence about 6.5% in healthy screened adults**, higher in cohorts with more men, older patients and smokers.
- MH is categorised by presence or absence of symptoms and quantified as RBCs per high-power field (HPF).

Concept — Timing within the stream localises the source, and microscopy — not colour, not dipstick — establishes that the red cells are real.

MICROSCOPIC HAEMATURIA

Criteria for Diagnosis of Microhaematuria

- **A few RBCs enter the urine normally**, and sexual activity and exercise can produce minor MH. Together with the high prevalence and low malignancy yield, this makes **a threshold with both high sensitivity and high specificity unattainable**.
- **Threshold ≥ 3 RBCs/HPF** is used by most studies and guideline panels.
- **Sensitivity–specificity trade-off** (large observational study correlating degree of MH with malignancy): 0–2 RBCs/HPF is fully sensitive but has no specificity; 3–10 gives moderate sensitivity and specificity; higher thresholds progressively lose sensitivity and gain specificity.
- The **American Urological Association (AUA)/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) guideline (2020)** defines MH as ≥ 3 RBCs/HPF: higher thresholds would miss treatable urological conditions.
- **A single positive UA is sufficient** to prompt evaluation, because MH from significant disease, including malignancy, can be **intermittent**.
- Other national guidelines vary (Table 8.1), reflecting different balances of benefit, harm and cost; **none has been robustly validated**.
- **Laboratory reporting must match the threshold**: a normal range extending above 3 (e.g. ≤ 5 RBCs/HPF) defeats it, and current reporting cut-offs are variable and often discordant with guideline risk groups.
- **Automated UA (e.g. flow cytometry)** increasingly replaces microscopy, but **the ≥ 3 RBCs/HPF threshold is based on microscopic sediment**; the correct automated threshold is unknown.

TABLE 8.1 SELECTED GUIDELINES FOR EVALUATION OF ASYMPTOMATIC MICROHEMATURIA

Guideline body	Population	Criterion for evaluation	Cystoscopy	Imaging
AUA/SUFU (2020); American Family Physician (2021) identical	Adults	≥ 3 RBCs/HPF on a single UA	Low risk optional; intermediate and high risk recommended	Low risk optional or renal ultrasound (RUS); intermediate RUS; high computed tomography (CT) urogram
American College of Physicians (2016)	Adults	≥ 3 RBCs/HPF on a single UA	Not stated	Not stated
Canadian Consensus Document (2016)	Adults	≥ 3 RBCs/HPF on 2 of 3 UAs	Age ≥ 35	Provider discretion
Kaiser Permanente (2009)	Adults	≥ 3 RBCs/HPF on 2 of 3 UAs	Urologist discretion	CT urogram or intravenous pyelogram (IVP); RUS
Dutch (2012)	Adults ≥ 50	≥ 3 RBCs/HPF on 2 of 3 UAs	Age ≥ 50 ; discretion if younger	Ultrasound if ≥ 50 ; discretion if younger
Japanese (2014)	Adults ≥ 40	≥ 5 RBCs/HPF on a single UA	Age ≥ 40 with risk factors	Ultrasound
American College of Obstetricians and Gynecologists (ACOG)/American Urogynecologic Society (AUGS) (2017, reaffirmed 2021)	Adult women	Not stated	Avoid in never-smoking women aged 35–50; evaluate only if ≥ 25 RBCs/HPF	

Guideline body	Population	Criterion for evaluation	Cystoscopy	Imaging
UK National Institute for Health and Care Excellence (NICE) (2016)	Age ≥60	≥1+ blood on dipstick PLUS dysuria or raised white cell count	Not stated	Not stated

Data from Barocas et al., 2020; Kassouf et al., 2016; Loo et al., 2009; ACOG Committee Opinion, 2017; Horie et al., 2014; van der Molen and Hovius, 2012; Nielsen and Qaseem, 2016; NICE, 2016; Arnold, 2021.

- **What the table shows:** guidelines diverge on the number of positive UAs, the RBC threshold, age floors and imaging modality.
- **Pattern:** North American bodies converge on ≥3 RBCs/HPF; **only AUA/SUFU (and American Family Physician) risk-stratify both cystoscopy and imaging;** most others default to ultrasound.
- **Traps:** NICE alone accepts a dipstick (with symptoms) — AUA/SUFU rejects dipstick diagnosis; AUA/SUFU needs one UA whereas Canadian, Kaiser and Dutch need 2 of 3; the Japanese threshold is ≥5; ACOG/AUGS withholds evaluation in never-smoking women 35–50 below 25 RBCs/HPF.

Differential Diagnosis

- **One-third to two-thirds of patients evaluated for MH have a demonstrable cause** — commonly benign prostatic enlargement (BPE), calculi and urethral stricture (Table 8.2).
- MH may arise anywhere: consider **neoplastic, infectious and anatomical causes** in kidney, ureter, bladder, prostate and urethra, plus **medical renal disease and mimics** such as vaginal bleeding.
- **Many benign findings need no treatment:** non-obstructing nephrolithiasis, pelvic organ prolapse, vaginal atrophy, BPE.
- **Malignancy is found in roughly 3% of MH evaluations,** varying with population, threshold and completeness of evaluation.
- Meta-analysis: **bladder cancer is found in a few per cent** (more in high-risk cohorts), whereas **upper tract urothelial carcinoma (UTUC) and renal cell carcinoma (RCC) are each found in about 1 in 1,000** — upper tract malignancy is very uncommon in MH.
- **Yield rises with degree of MH (>25 RBCs/HPF), GH and malignancy risk factors** (Box 8.1).

TABLE 8.2 DIFFERENTIAL DIAGNOSIS OF MICROHEMATURIA

Category	Example	Clinical presentation and risk factors
Neoplasm (risk factors: Box 8.1)	Bladder cancer	Older age, male, tobacco, occupational exposure, irritative voiding symptoms
	Ureteral or renal pelvis cancer	Family history of early colon, endometrial or upper tract tumours; Lynch syndrome
	Renal cortical tumour	Family history of early kidney tumours; flank pain or mass
	Prostate cancer	Older age, family history, Black race
	Urethral cancer	Obstructive symptoms, pain, bloody discharge
Infection and inflammation (history of infection)	Cystitis	Female predominance, dysuria
	Pyelonephritis	Fever, flank pain, diabetes, female predominance
	Urethritis	Sexually transmitted infection exposure, discharge, dysuria

Category	Example	Clinical presentation and risk factors
Calculus (prior stone)	Tuberculosis; schistosomiasis	Travel to endemic areas
	Haemorrhagic cystitis	See Box 8.2
	Nephroureterolithiasis	Flank or loin pain, family history
	Bladder stones	Bladder outlet obstruction
Benign prostatic enlargement	—	Male, older age, obstructive symptoms
Medical renal disease	Nephritis; immunoglobulin A (IgA) nephropathy	Hypertension, azotaemia, dysmorphic RBCs, cellular casts, proteinuria
Congenital or acquired anatomical abnormality	Polycystic kidney disease	Family history of renal cystic disease
	Pelviureteric junction obstruction	History of urinary tract infection (UTI), stone, flank pain
	Ureteral stricture	Prior surgery or radiation, flank pain, hydronephrosis
	Urethral stricture	Stranguria, spraying
	Urethral diverticulum	Discharge, dribbling, dyspareunia, UTI; female
	Fistula	Pneumaturia, faecaluria, abdominal pain, recurrent UTI, diverticulitis
Other	Exercise-induced haematuria	Recent vigorous exercise — diagnosis of exclusion
	Endometriosis	Cyclic haematuria in a menstruating woman
	Haematological or thrombotic disease	Personal or family history of bleeding or thrombosis
	Papillary necrosis	Black race, sickle cell disease, diabetes, analgesic abuse
	Arteriovenous malformation	Recent renal surgery (iatrogenic)
	Renal vein thrombosis; interstitial cystitis; trauma; recent surgery or instrumentation	Voiding symptoms (interstitial cystitis); relevant history

- **Footnotes:** the list assumes obvious benign causes (menstruation, recent instrumentation, uncomplicated cystitis) are excluded; **haematological illness, medical renal disease or anticoagulant/antiplatelet use does not preclude haematuria evaluation**; exercise-induced haematuria requires **confirmed absence of haematuria after abstaining from exercise**.
- **Purpose:** pairs each cause with the clinical clue that raises it. **Trap:** a plausible benign explanation is a hypothesis to be proven, not a reason to stop.

Bladder Cancer Detection and Risk Factors for Delayed Diagnosis

- United States: **over 82,000 new bladder cancers and over 16,700 deaths annually**; most new cases are non-muscle-invasive.
- **Male-to-female incidence 4:1, yet 5-year survival is worse in women** — partly from **delayed diagnosis**: women present at higher stage and their MH is more often **misattributed to UTI or gynaecological causes**.
- In muscle-invasive disease, **treatment delay worsens survival** → **prove rather than presume** a benign cause, and **repeat the UA after treatment** to confirm resolution.

- **Black patients: lower incidence but higher mortality**, likewise attributed to differences in referral and timely diagnosis.
- **Anticoagulants and antiplatelets** (warfarin, heparins, aspirin, clopidogrel, non-steroidal anti-inflammatory drugs): haematuria prevalence and cancer yield are **no different** from non-users; these drugs may **unmask lesions at an earlier stage**; a population study found users **more than twice as likely to be diagnosed with bladder cancer within 6 months of haematuria** → **risk-based evaluation exactly as for non-users**.
- **Under-evaluation is pervasive**: fewer than 25% of patients with haematuria are referred and fewer than 10% receive complete evaluation (cystoscopy and imaging), even when high risk.
- Contributors: **disparate guidelines**, low yield, cost of imaging and invasiveness of cystoscopy. Proposed remedies: **electronic record decision support, telemedicine consultation, one-stop haematuria clinics**.

BOX 8.1 UROTHELIAL CANCER RISK FACTORS

Included in AUA microhaematuria risk stratification	Additional urothelial cancer risk factors
Age	Irritative lower urinary tract symptoms
Male sex	Prior pelvic radiotherapy
Smoking	Prior cyclophosphamide or ifosfamide chemotherapy
Degree of MH	Family history of urothelial cancer or Lynch syndrome
Persistence of MH	Occupational benzene or aromatic amine exposure (rubber, petrochemicals, dyes)
History of MH	Chronic indwelling foreign body in the urinary tract

Not an exhaustive list. Used with permission from the American Urological Association.

- **Purpose**: the left column drives the risk stratum; the right column can lift a patient to **intermediate risk only** (Table 8.3). **Trap**: cyclophosphamide or radiotherapy exposure does not by itself make a patient high risk.

Rationale for a Risk-Based Evaluation Algorithm

- **One-third to two-thirds of MH evaluations are negative**; given cost and patient risk, an **individualised selection** for evaluation is sought.
- **Kaiser Permanente data**: patients with risk factors (age ≥ 50 , GH history, tobacco, male sex, >25 RBCs/HPF) had roughly ten-fold higher malignancy yield than intermediate-risk patients; low-risk yield was near zero.
- **ACOG/AUGS (2017)**: withhold evaluation in **never-smoking women aged 35–50 with <25 RBCs/HPF**, whose malignancy risk is **$<0.5\%$** .
- **AUA/SUFU (2020)** defined low, intermediate and high risk by **age, sex, smoking and degree of MH** (Table 8.3), **matching evaluation intensity to cancer risk**.
- Not externally validated at publication, but built on systematic review confirming each factor; later models use the same factors. **Subsequent validation** confirmed a stepwise rise in malignancy across the three strata.

TABLE 8.3 AMERICAN UROLOGICAL ASSOCIATION/SOCIETY OF URODYNAMICS, FEMALE PELVIC MEDICINE AND UROGENITAL RECONSTRUCTION RISK STRATIFICATION SYSTEM

Risk factor	Low (meets ALL)	Intermediate (meets ANY)	High (meets ANY)
Sex/age (years)	Women <50 ; men <40	Women 50–59; men 40–59	Women and men ≥ 60
Smoking (pack-years)	Never or <10	10–30	>30

Risk factor	Low (meets ALL)	Intermediate (meets ANY)	High (meets ANY)
Degree of MH (RBCs/HPF)	3–10	11–25	>25
History of GH	No	No	Yes
Prior episode of MH	No	Previously low risk, no prior evaluation, 3–25 RBCs/HPF on repeat UA	Previously low risk, no prior evaluation, >25 RBCs/HPF on repeat UA
Additional urothelial cancer risk factors (Box 8.1)	No	Yes	—
Malignancy risk (from text)	0–0.4%	1–2.5%	About 6%
Evaluation (Fig. 8.1)	Shared decision: repeat UA within 6 months or cystoscopy + RUS	Cystoscopy + RUS	Cystoscopy + CT urogram

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- **Asymmetry:** low risk requires **every** criterion; intermediate and high require **any one**. **Men enter intermediate a decade earlier than women** (40 vs 50), but both become high risk at 60.
- **Any history of GH → high risk** outright. Persistent MH re-stratifies a previously low-risk, unevaluated patient by the repeat count.

KEY POINTS: MICROSCOPIC HAEMATURIA

- **MH = ≥ 3 RBCs/HPF** on one or more urine microscopies; **dipstick is insufficient**.
- **Prevalence about 6.5%** of adults, varying by population.
- **Malignancy in about 3%** of MH evaluations — higher with more haematuria, male sex, tobacco and risk factors.

Concept — The diagnostic threshold is deliberately sensitive; specificity is recovered not by raising the threshold but by **stratifying the intensity of evaluation** to pre-test risk.

EVALUATION OF PATIENTS WITH MICROHAEMATURIA

- Many guidelines and risk models exist; the chapter follows the **AUA/SUFU guideline** (Fig. 8.1).

Initial Evaluation

- **History and examination focused on genitourinary malignancy risk factors** (Box 8.1), medical renal disease and benign causes: infection, trauma, nephrolithiasis, **BPE and urethral stricture in men, urethral diverticulum and pelvic organ prolapse in women**; directed examination, including female pelvic examination, as needed.
- **Measure blood pressure and serum creatinine** — both detect medical renal disease; creatinine also **guides the choice of imaging** where cross-sectional imaging is indicated.
- **Suspected benign cause → confirm it, treat, repeat UA**. Suspected UTI → **urine culture**, then follow-up UA after treatment; persistent MH → further evaluation. Omitting these steps delays bladder cancer diagnosis, raises stage and lowers cancer-specific survival.
- Benign conditions needing no treatment (non-obstructing stones, vaginal atrophy), or where treatment will not clear MH (medical therapy for BPE) → **shared decision-making** on further evaluation.

- Unprovable benign causes, such as **vigorous exercise**, are **diagnoses of exclusion**, not reasons to forgo evaluation.
- **Signs of medical renal disease** (hypertension, renal impairment, proteinuria, casts, dysmorphic RBCs) → **nephrology referral AND risk-based urological evaluation**, since bladder cancer may coexist.
- Then **assign low, intermediate or high risk** (Table 8.3).

FIGURE 8.1 AMERICAN UROLOGICAL ASSOCIATION GUIDELINE ALGORITHM FOR EVALUATION OF ADULT PATIENTS WITH MICROHEMATURIA

Sequence: MH ≥ 3 RBCs/HPF on UA with microscopy → history and examination (urothelial cancer risk factors, non-malignant causes) → suspected non-malignant or gynaecological source: directed evaluation (culture if infection suspected) → treat → repeat UA (negative → release) → risk stratification → evaluation by stratum (Table 8.3; CT-contraindicated high-risk patients receive MR urogram, or non-contrast imaging + retrograde pyelograms) → positive → treat; negative → consider repeat UA within 12 months (negative → release; positive → shared decision on repeat evaluation vs observation) → **re-evaluate if GH, increased MH or new urological symptoms**.

- **Figure footnotes:** low risk at presentation becomes intermediate or high only with **persistent MH**; consider a **catheterised specimen** in women with vaginal atrophy or prolapse; untreated chronic benign conditions need judgement and patient engagement; **cross-sectional or retrograde imaging if MH persists after a negative RUS**.

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Low-Risk Patients With Microhaematuria

- **Malignancy risk 0–0.4%. More than 85% of malignancies found on MH evaluation are bladder cancers;** UTUC and RCC are uncommon overall and particularly rare in low risk.
- Low yield against cost, false positives, complications and discomfort makes benefit debatable; **the purpose of stratification is largely to identify a group safe to forgo evaluation**.
- **Shared decision-making: repeat UA within 6 months vs cystoscopy + RUS**, discussing low yield, false positives, cystoscopy complications, discomfort, cost, preference and risk tolerance.
- **Repeat UA still positive → cystoscopy and imaging**, with modality set by the new stratum (3–25 RBCs/HPF intermediate; >25 high), assuming no new irritative symptoms.
- Omitting evaluation in some and substituting RUS for CT urography **reduces burden, risk and cost without compromising cancer detection**.

Intermediate-Risk Patients With Microhaematuria

- **Malignancy risk 1–2.5%;** UTUC risk remains very low → **cystoscopy + RUS**.
- **RUS retains sensitivity for renal cortical tumours** while avoiding ionising radiation and iodinated contrast, and reduces incidental findings and cost.

High-Risk Patients With Microhaematuria

- **Malignancy about 6%** in validation, higher in men, smokers and those with GH history.
- UTUC is uncommon overall but **likely most frequent in this stratum**; many guidelines use RUS for all, but **RUS is poorly sensitive for UTUC** → AUA/SUFU: **cystoscopy + axial upper tract imaging with excretory views**.

Upper Tract Imaging in the Diagnostic Evaluation of Microscopic Haematuria

- **CT urography has the highest sensitivity and specificity** for renal parenchymal and upper tract lesions, is very sensitive for calculi and shows extra-urinary disease.

- **Limitations:** **pregnancy, iodinated contrast allergy, chronic kidney disease; radiation 20–30 mSv** (variable between centres), with potential carcinogenesis.
- **Cost-effectiveness modelling:** CT urography instead of RUS costs **over US\$1 million per additional cancer detected**, with very few extra cancers, because UTUC is rare; across large populations, radiation-induced cancers, contrast complications and false positives become substantial.
- Some organisations therefore use RUS for all; AUA/SUFU instead reserves **axial imaging for the high-risk subpopulation**, whose upper tract cancer risk justifies it.
- **CT urogram phases:** pre-contrast (stones, hyperdense lesions) → contrast-enhanced parenchymal (cortical tumours) → delayed excretory images from kidneys to bladder (urothelium).
- **RUS sensitivity for UTUC is poorly defined but appears lower than CT** — only two studies, with conflicting results; CT urography shows high sensitivity and very high specificity for UTUC and RCC combined.
- **CT risk mitigation:** split-bolus protocols (three series reduced to two) and radiation dose adjusted for body mass index.
- **Contrast-associated acute kidney injury (AKI):** an American College of Radiology (ACR)/National Kidney Foundation consensus states the risk has been **overstated**, recommending hydration when **glomerular filtration rate (GFR) <30 mL/min/1.73 m²**.
- **CT urogram contraindicated → MR urography;** MRI contraindicated (some pacemakers, metal implants, severe claustrophobia) → **non-contrast CT or ultrasound of the parenchyma + retrograde pyelography** of calyces, pelvis and ureters.
- **Nephrogenic systemic fibrosis (NSF)** after gadolinium is small and falling: **exceedingly rare at GFR ≥30;** below that, **group II or III gadolinium agents** may be used.
- **Genetic predisposition to renal cortical tumours (e.g. von Hippel-Lindau disease) → upper tract imaging regardless of stratum.**
- **Persistent or recurrent MH after negative RUS** (low or intermediate risk) → cross-sectional or retrograde urography may be used.

COMPARISON — CT UROGRAPHY VS RENAL ULTRASOUND IN MICROHAEMATURIA

Attribute	CT urography	Renal ultrasound
Sensitivity for UTUC	High	Low, poorly defined
Renal cortical tumours	High	Sensitivity retained
Calculi and extra-urinary disease	Detected	Limited
Radiation and contrast	20–30 mSv; iodinated contrast	None
Incidental findings and cost	More	Fewer
AUA/SUFU use	High risk	Intermediate risk; low risk opting for evaluation

Urine Cytology and Urinary Biomarkers in the Evaluation of Haematuria

- **Cytology and biomarkers are not recommended in the initial evaluation of MH:** none has shown incremental value over cystoscopy and imaging in a pure MH population; supportive studies are in bladder cancer surveillance or mixed GH/MH cohorts.
- **Cytology:** moderate sensitivity, high specificity in a mixed cohort, yet **no cancer was found in patients with positive cytology but normal cystoscopy and upper tract imaging;** in a large prospective MH

series, positive cytology with a negative work-up rarely yielded later cancer, and false positives were appreciable.

- US Food and Drug Administration-approved or cleared assays have **rarely been studied in MH without prior bladder cancer**:

COMPARISON — URINARY BIOMARKER ASSAYS

Assay	Principle and targets
Nuclear matrix protein 22 (NMP22)	Protein; qualitative and quantitative
Bladder tumour antigen (BTA)	Protein; qualitative and quantitative
UroVysion	Fluorescence in situ hybridisation (FISH): chromosomes 3, 7, 17 and 9p21
ImmunoCyt	Immunocytology: carcinoembryonic antigen and mucin glycoproteins
Xpert Bladder	Messenger ribonucleic acid (mRNA): UPK1B, IGF2, CRH, ANXA10, ABL1
Cxbladder	Reverse transcription polymerase chain reaction (RT-PCR), five mRNAs: IGFBP5, HOXA13, MDK, CDK1 (CDC2), CXCR2
AssureMDx	Deoxyribonucleic acid (DNA): FGFR3, TERT, HRAS mutations; methylation of OTX1, ONECUT2, TWIST1

- Pooled sensitivities and specificities are moderate to good; **no assay is shown superior**, and most data are from mixed populations.
- **Likelihood-ratio modelling**: in a low-risk patient, a negative test lowers cancer risk to negligible and a positive test raises it only modestly.
- **None is sensitive or validated enough to replace cystoscopy or imaging. Cytology may be considered after a negative work-up if urothelial carcinoma is still suspected**, particularly with irritative voiding symptoms.

Natural History of Microhaematuria in Patients With a Negative Initial Evaluation

- **MH resolves in about one-third** over months to years — but these cohorts were young and often incompletely evaluated.
- Re-evaluation series found a small malignancy yield, but **initial evaluations were often incomplete** and most cancers came from CT in patients not previously CT-imaged.
- After a **complete** initial evaluation, **malignancy on a second evaluation is rare**; long-term follow-up of men over 50 found only occasional bladder cancers, many years later.
- AUA/SUFU (conditional): **may repeat UA within 12 months and discontinue if negative**.
- **Persistent or recurrent MH → shared decision** on re-evaluation, acknowledging low yield.
- **GH, a significant rise in MH or new urological symptoms → re-evaluate**: these suggest new or occult disease.
- **Cytology** if carcinoma in situ risk factors develop (GH, irritative symptoms); **cross-sectional or retrograde urography** if initially imaged with RUS.
- Untreated chronic benign causes (non-obstructing stones, BPE, prolapse) → judgement and shared decision on whether and when to re-evaluate.
- **Low-risk patients who skipped evaluation → risk-based evaluation if MH persists or recurs**.

KEY POINTS: EVALUATION OF PATIENTS WITH MICROHAEMATURIA

- **History, examination, renal function and blood pressure → risk stratification → stratum-matched evaluation (Table 8.3).**

- **White-light cystoscopy** rather than enhanced cystoscopy; no cytology or biomarkers initially.
- **Negative complete evaluation → release** once UA confirms resolution; re-evaluation considered for persistent/recurrent MH or incomplete initial evaluation.

Concept — Bladder cancer dominates the yield, so cystoscopy is near-universal; upper tract cancer is rare, so CT urography is reserved for the stratum in which the rare becomes worth the radiation, contrast and cost.

GROSS HAEMATURIA

- The differential is the same as for MH, but the yield of significant lesions rises with the degree of haematuria: about half of GH patients have a demonstrable cause.
- **DETECT I** (prospective, UK, 40 hospitals, 3,556 referred patients): urinary tract cancer in **13.8% with GH** vs 3.1% with MH; 3.5% of GH patients under 45 had cancer → investigate GH without an age threshold.
- **IDENTIFY** (international prospective, 10,896 patients, 110 hospitals, 26 countries): cancer in **26% with GH** vs 6.38% with MH.
- In both, **bladder cancer was the commonest cancer**; a systematic review likewise found bladder cancer, RCC and UTUC all several-fold more frequent with GH than MH.
- Risk prediction models from DETECT and IDENTIFY both retain **GH as a significant predictor of cancer**.
- **GH without trauma or culture-documented UTI → urine cytology, cystoscopy and upper tract imaging, preferably CT urography.**
- **GH with culture-documented UTI → treat, then repeat UA** to confirm clearance.
- Initial assessment as for MH, plus: **haemodynamic stability, full blood count for anaemia, coagulation parameters if anticoagulated; bladder emptying (post-void residual ultrasound)**, with catheterisation and clot irrigation if clot retention is suspected.
- Stabilise → diagnostic evaluation → cause-specific management.
- **Non-diagnostic evaluation:** follow the MH schedule as a reference; **full repeat evaluation if GH recurs.**

COMPARISON — EVALUATION OF MICROHAEMATURIA VS GROSS HAEMATURIA

Attribute	Microhaematuria	Gross haematuria
Cancer yield	About 3%	Several-fold higher
Risk stratification	Yes — sets intensity	No — uniform full evaluation, no age threshold
Cystoscopy	Optional (low), recommended (intermediate, high)	All
Upper tract imaging	RUS or CT urogram by stratum	CT urography preferred
Urine cytology	Not recommended initially	Recommended
Acute assessment	—	Haemodynamics, full blood count, coagulation, post-void residual, clot evacuation

Concept — GH carries enough cancer risk that **stratification is abandoned**: every patient receives the full triad of cytology, cystoscopy and CT urography, at any age.

HAEMORRHAGIC CYSTITIS

- **Spectrum from transient to life-threatening**, often in elderly, infirm, comorbid patients → **central tenet: weigh the risk of the current state against the risk of escalating treatment.**
- **Diffuse inflammation and bleeding of the bladder mucosa**; causes in Box 8.2.
- **Bacterial infection**: a common cause of GH; resolves with appropriate treatment.
- **Viral**: children and immunosuppressed adults, especially after **renal or bone marrow transplantation**. **BK virus (polyomavirus) is the commonest**; **adenovirus types 11 and 35** in children and renal transplant recipients. Treatment is **supportive** (hydration, diuresis, bladder irrigation); antiviral success is anecdotal.
- **Oxazaphosphorines (cyclophosphamide, ifosfamide)**: frequent and **dose-dependent**. Liver produces **acrolein** → renal excretion → urothelial contact → mucosal sloughing → oedema and fibrosis. **Onset typically within 48 hours.**
- **Mesna (2-mercaptoethane sulfonate) binds acrolein and inactivates it** as prophylaxis, yet a substantial minority still develop cystitis; **whether mesna beats hyperhydration with forced diuresis and/or continuous bladder irrigation (CBI) is unresolved.**
- **Pelvic radiotherapy**: moderate-to-severe haematuria in about 5%, dose- and technique-dependent, **onset 6 months to 10 years** after treatment; reported incidence varies with population and follow-up, rising with long follow-up after post-prostatectomy radiotherapy, and **anticoagulant/antiplatelet use increases risk.**
- **Radiation mechanism**: vascular endothelial damage → inflammation → fibrosis → ischaemia → **progressive obliterative endarteritis** → necrosis and mucosal sloughing; **secondary infection** further impairs healing.
- **Morbidity is substantial**: repeated admissions, transfusion, cystoscopic fulguration and clot evacuation, and some deaths.
- **Cyclophosphamide and pelvic radiation are bladder cancer risk factors → exclude bladder cancer before labelling haemorrhagic cystitis.**

BOX 8.2 DIFFERENTIAL DIAGNOSIS FOR HEMORRHAGIC CYSTITIS

Category	Causes
Infectious	Bacterial; viral (especially BK virus, adenovirus); fungal; parasitic
Trauma	External; postsurgical (e.g. transurethral resection of bladder)
Malignancy	Bladder primary; invasion from local or distant primary
Vascular malformation	—
Chemical exposure	Cyclophosphamide, ifosfamide, busulfan, thiotepa, temozolomide, aniline dye, ether, nonoxynol-9 (accidental urethral insertion of vaginal contraceptive)
Radiation therapy	E.g. prostate or cervical cancer
Medication-induced	Penicillins (immune reaction), bleomycin, danazol, tiaprofenic acid, allopurinol, phensuximide, methenamine mandelate, acetic acid
Systemic disease	Amyloidosis, rheumatoid arthritis, Crohn disease

- **Definition implied by the box**: bleeding localised to the bladder **after full GH work-up (cystoscopy, cytology, upper tract imaging)** without another source. **Trap**: chemical causes extend well beyond the oxazaphosphorines.

Management of Haemorrhagic Cystitis

- Cause-directed treatment (e.g. of infection) is occasionally possible; usually a **sequential, severity-guided escalation** is followed (Fig. 8.2).
- **First line (usually sufficient for mild cases):** hydration/diuresis, catheter with CBI, transfusion as needed.
- **Persistent bleeding or uncontrollable clot → cystoscopy under anaesthesia:** clot evacuation, fulguration of discrete bleeding points, **biopsy of any suspicious area**.
- **No large prospective comparative trials** of intravesical agents exist:

COMPARISON — INTRAVESICAL AGENTS FOR HAEMORRHAGIC CYSTITIS

Agent	Mechanism	Administration	Efficacy	Key precaution
Alum (aluminium ammonium or potassium sulfate)	Astringent: protein precipitation on urothelium → vasoconstriction, ↓ capillary permeability	50 g in 5 L sterile water (1%), irrigate 200–300 mL/h; no anaesthesia	Widely variable; limited durability	Low toxicity (suprapubic pain, spasms); aluminium toxicity with encephalopathy in renal insufficiency
Prostaglandins (carboprost, prostaglandin F2α)	Vasoconstriction, platelet aggregation, cytoprotection via mucus barrier	Intravesical	Moderate; no difference from alum in a small randomised trial	Access, storage, cost
Sodium hyaluronate	Restores glycosaminoglycan layer	Intravesical	Symptomatic improvement	—
Silver nitrate	Chemical coagulation	0.5–1% under gravity, 10–20 min	—	Precipitation → upper tract obstruction → cystogram to exclude reflux
Aminocaproic acid	Lysine analogue; competitive inhibitor of plasminogen activators (incl. urokinase) → blocks fibrinolysis	CBI 200 mg/L in 0.9% saline, continued 24 h after bleeding stops	High resolution reported	Thromboembolism; only in a clot-free bladder — otherwise hard, organised clots
Formalin	Protein precipitation, capillary occlusion	1–4% ; under general or spinal anaesthesia; see sequence below	Highest (80–90%)	Pain; bladder fibrosis, ↓ capacity; ureteric stricture, hydronephrosis; pre-treatment cystogram

- **Alum is first-line intravesical therapy** after failed supportive care, **unless renal insufficiency**.
- **Formalin sequence:** cystogram to exclude reflux or perforation (reflux → **occlusive ureteric catheters**) → anaesthesia → instil **up to 300 mL or bladder capacity** under gravity, catheter **≤15 cm above the pubic symphysis** → dwell **10–15 min** with the catheter on light traction (protects urethra) and external skin protected. **Use the lowest concentration:** efficacy and complications both rise with dose. **Reserved as second-line** given toxicity and the need for anaesthesia.

Hyperbaric Oxygen Therapy

- **Clinically stable patients, particularly after radiation or cyclophosphamide** → hyperbaric oxygen therapy (HBOT) is the next step.
- **100% oxygen at 2–3 atmospheres, about 90 min, 30–40 sessions** → ↑ tissue oxygen tension → ↓ oedema, neovascularisation, wound healing.

- **Response 80–90%**, sustained up to about 2.5 years. **RICH-ART** randomised trial (radiation cystitis): HBOT significantly improved the Expanded Prostate Cancer Index Composite (EPIC) urinary score (about 10 points) vs standard care.
- **Durability is limited**: most relapse with longer follow-up, with only about a quarter in complete response at 5 years.
- Complications: **claustrophobia, otalgia, barotrauma, myopia, rarely seizures**.
- Continued bleeding during HBOT → **bilateral nephrostomies ± ureteric occlusion**, removing urokinase-containing urine from the bladder to aid haemostasis over the long HBOT course.

Angioembolisation

- **Clinically unstable patients or intractable bleeding** → **internal iliac artery angioembolisation** (first reported 1974); unilateral or bilateral, low risk even in debilitated patients.
- **Selective bilateral embolisation of the anterior division** is usually needed (gelatin microspheres, polyvinyl alcohol particles, coils). **Avoid the posterior division** → superior gluteal artery occlusion → **gluteal pain**.
- Initial control about 90%, but **durability unknown**; series are small and include pelvic tumours; reliability in **radiation cystitis, where mucosal ischaemia is the pathology**, is uncertain.

Cystectomy and Urinary Diversion

- Failed non-surgical management → **cystectomy with urinary diversion**.
- **Supravescical diversion (e.g. ileal conduit) without cystectomy**: complications, chiefly **pyocystitis**, and frequent readmission → **remove the bladder at diversion if feasible**.
- Patients are elderly and deconditioned → **perioperative complications higher than cystectomy for cancer**; yet definitive surgery reduces subsequent admissions, procedures and transfusions → **balance surgical risk against deconditioning from prolonged conservative care**.

Emerging Options

- **Tranexamic acid** (antifibrinolytic): intravenous use lowers transfusion at radical cystectomy; benefit, route and **risk of clot retention** in haemorrhagic cystitis unestablished. **Laser ablation** of mucosal bleeding: feasible, needs study.

FIGURE 8.2 MANAGEMENT ALGORITHM FOR PATIENTS WITH HEMORRHAGIC CYSTITIS

Diagnose (risk factors; evaluation with upper tract imaging, cytology, culture) → **initial management**: hydration/diuresis, cystoscopy with clot evacuation ± fulguration (biopsy if suspicious), CBI, transfusion as needed, correct infection, coagulopathy and tumour, **intravesical alum** → bleeding after 48 h → **HBOT** with supportive care; bilateral nephrostomies ± nephro-occlusion if bleeding continues → bleeding after HBOT → **formalin 1%** (pre-treatment cystogram) → bleeding after 48 h → **formalin 4%** (repeat cystogram) → continued bleeding → **cystectomy with conduit diversion**.

KEY POINTS: HAEMORRHAGIC CYSTITIS

- **Oxazaphosphorines cause cystitis via urothelial exposure to acrolein**.
- **HBOT achieves high response rates**.
- **Formalin is highly effective; obtain a cystogram first to exclude vesicoureteric reflux**.

- **Weigh cystectomy morbidity against timely definitive intervention** to prevent deconditioning from prolonged conservative efforts.

Concept — Management escalates from least to most morbid, with **clinical stability choosing the branch**: stable patients buy time with HBOT, unstable patients go to embolisation, and cystectomy is weighed against the deconditioning each delay causes.

HAEMATURIA FROM PROSTATE ORIGIN

- **A diagnosis of exclusion after complete GH evaluation** (cytology, upper tract imaging, cystoscopy). Severity ranges from self-limiting episodes to obstructing, transfusion-dependent bleeding.
- Causes: **benign prostatic hyperplasia (BPH), prostatitis, prostate cancer** (Fig. 8.3).

Benign Prostatic Hyperplasia

- **Commonest cause of prostatic bleeding and of GH in men over 60**; the only pathology found in about one-fifth of haematuria series.
- **Mechanism: increased microvessel density** in hyperplastic tissue, driven by **vascular endothelial growth factor (VEGF)**.
- Usually mild and self-limiting → **expectant management with hydration**.
- GH was historically a surgical indication; understanding the VEGF pathway has introduced **targeted medical therapy**.
- **Androgen suppression** (oestrogens, antiandrogens; case reports) represses androgen-driven angiogenesis and induces apoptosis.
- **Finasteride** (5 α -reductase inhibitor (5-ARI); blocks testosterone → dihydrotestosterone) ↓ VEGF expression, microvessel density and prostatic blood flow; **improvement or resolution in about 90%**, including anticoagulated patients. A randomised trial vs expectant management showed **markedly fewer recurrences and no surgery for bleeding with finasteride. Onset 2 weeks to 9 months**.
- Randomised trial: **finasteride and cyproterone acetate equally reduce recurrence** vs watchful waiting. **Dutasteride** (type I and II inhibition) produces similar microvessel changes.
- **5-ARIs have the best data and fewest side effects. AUA BPH guideline (2021, Expert Opinion)**: 5-ARIs appropriate for **refractory prostatic haematuria after excluding other causes**, and may be considered to **reduce intraoperative bleeding and transfusion at transurethral resection of the prostate (TURP) or other BPH surgery** (randomised data conflicting).
- **Clot or retention** → large-bore catheter, complete clot evacuation → CBI until clear → failure → **endoscopy under anaesthesia**: clot evacuation, electrocautery or laser.
- Intravesical agents (e.g. aminocaproic acid) have **little supporting evidence** here.
- **Refractory bleeding → TURP**, particularly if other surgical indications coexist; alternatives include photoselective vaporisation of the prostate (PVP), holmium laser enucleation of the prostate (HoLEP) and open simple prostatectomy — **the principle is removal of friable hyperplastic transition-zone tissue**.
- **Bleeding despite TURP → prostatic artery embolisation (PAE)**: bilateral, preferably superselective with gelatin microparticles (or anterior division of internal iliac); high initial success, bleeding stops within days, durable at 12 months. AUA 2021 cites PAE as an adjunct for refractory haematuria.
- Radical prostatectomy or cystoprostatectomy: last resort in usually poor candidates.

Prostatitis

- Haematuria presents in a small minority; mechanism unclear, probably inflammatory. **Culture-proven bacterial prostatitis → antibiotics**; significant recurrent haematuria in non-bacterial prostatitis is uncommon — antibiotics plus supportive care suggested.

Prostate Cancer

- Bleeding usually reflects **locally advanced disease with bladder base or trigonal invasion (T4)**; **haematuria is the commonest local symptom of advanced symptomatic prostate cancer**.
- **Confirm the prostatic source endoscopically**, especially after radiotherapy — exclude radiation cystitis or second bladder malignancy.
- Patients are elderly and unwell → **palliative intent**. Mild bleeding → catheter ± CBI.
- Not acutely life-threatening, radiation-naïve → **palliative external beam radiotherapy (EBRT) ± androgen deprivation therapy (ADT)**: good early response, **poor durability**.
- Unfit for local therapy or recurrent after local therapy → **ADT**, which reduces prostatic vascularity.
- Persistent bleeding, especially with bladder outlet obstruction → **cystoscopy with fulguration and/or limited ("channel") TURP**; selective angioembolisation possible but scant data and limited durability.
- Escalating bleeding → **urinary diversion, first percutaneous nephrostomy** → then **palliative extirpation**, usually **cystoprostatectomy with conduit**, depending on status.

FIGURE 8.3 MANAGEMENT ALGORITHM FOR PERSISTENT HEMATURIA OF PROSTATE ORIGIN

Cause	First line	Escalation	Salvage
All (common start)	Exclude other sources (upper tract imaging, cytology, cystoscopy) → hydration/diuresis, CBI, transfusion as needed, correct coagulopathy		
BPH	5-ARI	Cystoscopy with fulguration; consider antiandrogen	Poor clinical status → angioembolisation ; fit → BPH surgery (TURP, HoLEP, PVP, simple prostatectomy)
Prostatitis	Antibiotics (diagnosis by presentation + urine culture)	—	—
Prostate cancer (usually advanced)	ADT ± EBRT	Cystoscopy with fulguration / channel TURP; angioembolisation may be considered	Urinary diversion : nephrostomy → palliative prostatectomy or cystoprostatectomy with conduit

KEY POINTS: HAEMATURIA FROM PROSTATIC ORIGIN

- **BPH is the commonest cause of GH in men over 60.**
- **5-ARIs** treat BPH-related GH, with improvement within weeks.
- **ADT** may control GH from locally advanced prostate cancer.
- **Angioembolisation and/or urinary diversion** are salvage options, depending on status.

Concept — Prostatic bleeding is vascular, so its treatment is anti-vascular: **5-ARIs and ADT shrink the VEGF-driven microvasculature, surgery removes the friable tissue, and embolisation cuts the supply.**

URETHRAL BLEEDING

- **Urethrorrhagia**: bleeding from the urethra distal to the bladder neck, independent of micturition.

- **Blood at the meatus without voiding, or initial haematuria → pathology distal to the external sphincter.** In women, history rarely separates urethral from gynaecological bleeding → **pelvic examination**.
- **Retrograde urethrogram (RUG) and cystourethroscopy are the diagnostic mainstays:** visualisation, biopsy and fulguration.
- Causes are **classified by sex** (Box 8.3).
- **Men: trauma is the commonest cause.** Blunt injury (straddle, perineal kick, pelvic fracture) often presents with bleeding and **retention**; perineal or penile bruising with haematoma indicates injury; **RUG is essential when urethral injury is suspected**.
- **Foreign body insertion** → imaging and/or cystoscopy to exclude retained fragments, which perpetuate bleeding or form calculi.
- **Penile fracture with blood at the meatus or haematuria → prompt RUG or cystoscopy before exploration** to define the site and nature of urethral injury.
- **Urethritis:** bacterial or viral infection, chemical irritants (spermicidal jelly), drugs (amiodarone), rarely autoimmune (human leucocyte antigen B27 (HLA-B27) reactive arthritis); urethral discharge on palpation in men; investigate with **urine microscopy, culture and urethral swabs**.
- **Urethral tumours are rare;** blood per meatus is classic for **urothelial carcinoma in a urethra retained after radical cystectomy**.
- **Urethral caruncle:** benign, from the **posterior lip**, in **postmenopausal women**; distal urethral prolapse due to **oestrogen deficiency**.
- **Urethral diverticulum:** classic dysuria, dyspareunia and dribbling, plus **intermittent bleeding** and discharge on examination.

BOX 8.3 DIFFERENTIAL DIAGNOSIS FOR URETHRAL BLEEDING

Male	Female
Trauma: blunt (straddle, perineal kick); penetrating (foreign body, failed catheterisation, ballistic); intercourse-related (penile fracture, masturbation)	Trauma: blunt (pelvic fracture); penetrating (foreign body)
Urethritis: bacterial (gonococcal, non-gonococcal), viral, chemical (amiodarone), autoimmune (reactive arthritis)	Urethritis
Malignancy: urothelial carcinoma; squamous cell carcinoma (meatus, glans); condyloma	Malignancy
Calculus disease	Calculus disease
—	Urethral diverticulum; urethral caruncle

- **Pattern:** the male list is dominated by trauma subtypes; **diverticulum and caruncle are female-only**.

Concept — Bleeding independent of voiding places the lesion **distal to the sphincter**; the differential then divides by sex, and suspected trauma always means **RUG before instrumentation or exploration**.

HAEMATURIA ORIGINATING FROM THE UPPER URINARY TRACT

- Often asymptomatic; heavy bleeding with clots causes ureteric obstruction (**clot colic**), anaemia and rarely instability.
- Typically **total haematuria**, sometimes with **worm-like clots**.
- **Commonest causes: stones, trauma and malignancy** (covered elsewhere); the chapter highlights less frequent causes (Box 8.4).

BOX 8.4 DIFFERENTIAL DIAGNOSIS FOR UPPER URINARY TRACT BLEEDING

Category	Causes
Glomerular	IgA nephropathy (Berger disease), thin basement membrane disease, acute (e.g. post-streptococcal) glomerulonephritis, lupus nephritis, hereditary nephritis (e.g. Alport syndrome)
Tubulointerstitial	Papillary necrosis, sickle cell nephropathy, analgesic nephropathy, polycystic kidney disease, medullary sponge kidney
Vasculitis	Henoch–Schönlein purpura (IgA vasculitis); granulomatosis with polyangiitis (formerly Wegener)
Infection	Pyelonephritis, xanthogranulomatous pyelonephritis, renal tuberculosis, fungal infection
Obstruction	Pelviureteric junction obstruction, ureteral stricture, nephrolithiasis
Malignancy	Renal cortical tumours (RCC, benign), UTUC, fibroepithelial polyp
Vascular	Renal arteriovenous malformation (AVM) (congenital, acquired), iliac arteriovenous fistula, renal artery aneurysm (especially ruptured), pseudoaneurysm, renal artery/vein thrombosis, haemangioma, atheroembolism (including infarction), nutcracker syndrome, loin pain–haematuria syndrome
Trauma; other	Blunt, penetrating; lateralising essential haematuria

Medical Renal Disease

- **Glomerular diseases** (acquired or inherited) → haematuria, proteinuria with hypoproteinaemia and oedema, reduced GFR. **RBC casts, dysmorphic RBCs and proteinuria indicate a glomerular source.**
- **Tubulointerstitial diseases** affect structures outside the glomerulus. **Sickle cell nephropathy:** sickled cells ↓ medullary flow → ischaemia, microinfarction → **papillary necrosis**. **Analgesic nephropathy** → papillary necrosis → chronic interstitial nephritis.
- **Percutaneous renal biopsy** when glomerular or tubulointerstitial disease is suspected.
- **Complete risk-stratified haematuria evaluation even when medical renal disease is suspected** and nephrology referral is made.

Vascular Conditions Affecting the Urinary Tract

- **Ureteroiliac artery fistula:** rare, **life-threatening**; predisposed by pelvic or vascular surgery, pelvic irradiation, extensive ureteric mobilisation and **chronic ureteric stenting**. Open repair carries high mortality → **angiographic localisation with endovascular stenting preferred.**
- **Renal AVM:** congenital or acquired; **acquired forms are 75%** (renal biopsy, partial nephrectomy, nephrolithotomy, trauma). **Arteriography with selective angioembolisation** is the primary diagnostic and therapeutic step, preserving parenchyma; aim is occlusion of the arteriovenous communication. **GH after a recent renal procedure → expeditious angiography.**
- **Renal artery aneurysm:** linked to connective tissue disorders, usually asymptomatic; **hypertension in up to 90%**; dissection → flank pain and GH. Aneurysm and pseudoaneurysm: **blood pressure control → endovascular repair if refractory but stable → surgery if unstable.**
- **Nutcracker syndrome** (renal vein entrapment): **left renal vein compressed between aorta (posterior) and superior mesenteric artery (anterior)** → ↑ left renal vein pressure → rupture of thin-walled capillaries into the collecting system. Surgery: left renal vein transposition, superior mesenteric artery transposition, nephrectomy; **endovascular stenting** more recently.

COMPARISON — VASCULAR CAUSES OF UPPER TRACT BLEEDING

Entity	Key predisposition	Preferred management
Ureteroiliac fistula	Pelvic/vascular surgery, radiation, ureteric mobilisation, chronic stent	Endovascular stenting (open repair high mortality)

Entity	Key predisposition	Preferred management
Renal AVM	Mostly acquired: biopsy, partial nephrectomy, nephrolithotomy, trauma	Selective angioembolisation
Renal artery aneurysm/ pseudoaneurysm	Connective tissue disorders	Blood pressure control → endovascular if stable → surgery if unstable
Nutcracker syndrome	Aortomesenteric left renal vein compression	Vein or artery transposition, nephrectomy, renal vein stenting

Lateralising Essential Haematuria and the Evaluation of Upper Urinary Tract Bleeding

- Also **benign essential** or **chronic unilateral essential haematuria: macroscopic haematuria** **cystoscopically lateralised to one side with no identifiable cause**, prior imaging normal; rare; ranges from mild GH to clot retention and anaemia.
- **Cystoscopy during active bleeding lateralises the source** → **ureteropyeloscopy** is diagnostic and potentially therapeutic.
- **Technique:** judicious guidewire use (avoid urothelial trauma that mimics lesions) → **low-pressure irrigation** → **systematic inspection of every calyx, superior to inferior** → biopsy suspicious lesions → fulgurate tumours or bleeding sources (e.g. haemangioma).

KEY POINTS: URETHRAL BLEEDING AND HAEMATURIA ORIGINATING FROM THE UPPER URINARY TRACT

- **Suspect urethral bleeding** with blood at the meatus and/or initial haematuria; **suspected traumatic injury** → RUG.
- **RBC casts, dysmorphic RBCs and proteinuria** suggest medical renal disease; **risk-based evaluation is still performed**.
- **GH after a recent renal procedure** → **expeditious angiography** for renal AVM.

Concept — For vascular causes of upper tract bleeding, **angiography is both the test and the treatment**; open surgery is reserved for instability or failure of the endovascular route.